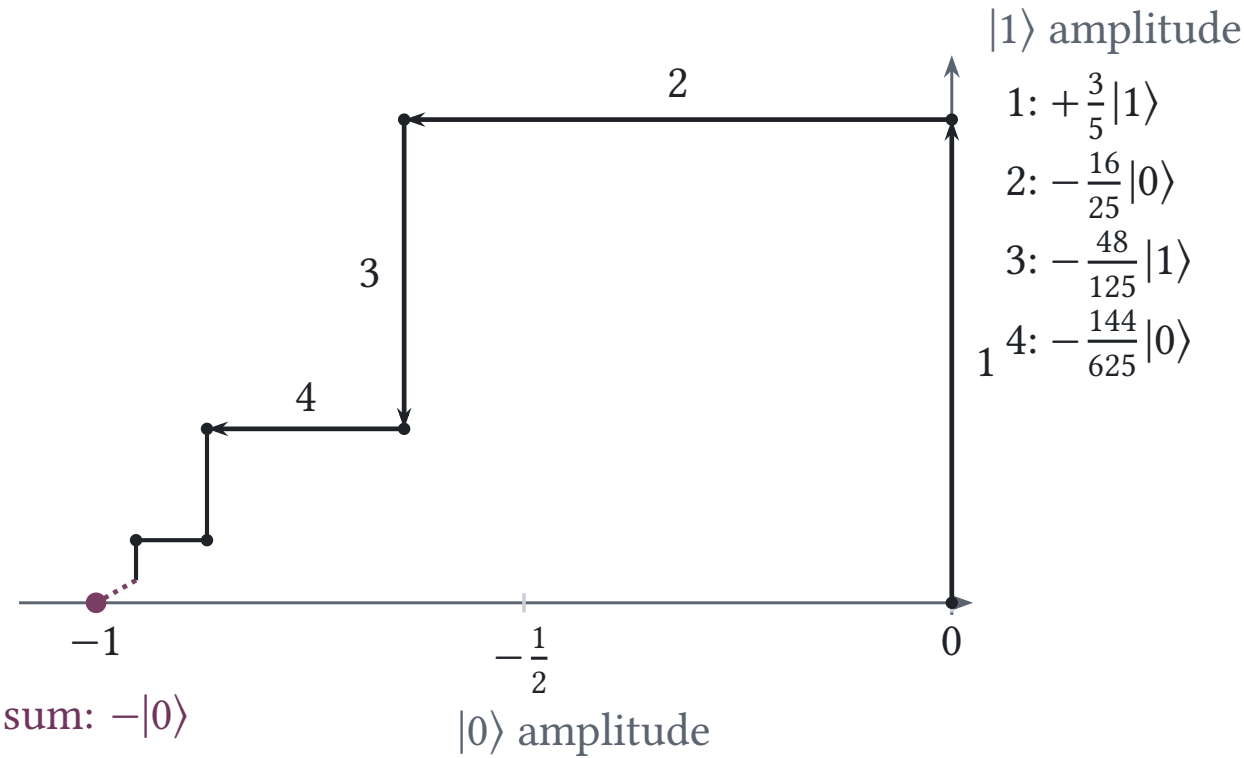
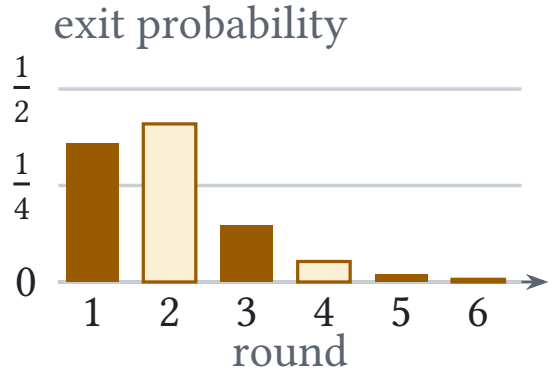
(a) `trace`*: add the exit vectors*



each arrow adds one exit
intermediate points are unnormalized sums

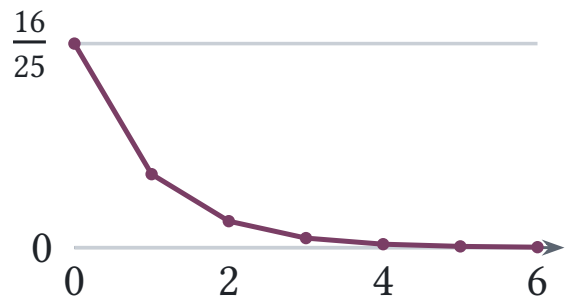
(b) `while`*: add exit probabilities*



odd total: 9/17 on $|1\rangle$

even total: 8/17 on $|0\rangle$

(c) probability still inside



after $k+1$ rounds
each further round retains 9/25